West Pharmaceutical Services, Inc.
347 Oliver Street
Jersey Shore, PA 17740
www.westpharma.com

QUALITY CERTIFICATE

No. 80372784 - 000010

Compliance Statements

West Pharmaceutical Services, Inc. (West) certifies that all the lots of material listed on this certification have been manufactured according to current Good Manufacturing Practices and are made in accordance to the referenced specification along with any conditions that may be noted in the customer specification review. If a customer specification has not been approved by West, the lots of material will be manufactured, inspected, and packaged to the previously agreed customer specification. If a customer specification has not been provided, the applicable West Master Specification will apply.

The formulation complies with the chemical requirements in the corresponding monograph 3.2.9 for "rubber closures" Type I in the valid version of the European Pharmacopoeia.
The formulation complies with the chemical requirements of the Japanese Pharmacopoeia, current edition, chapter 7.03.
The formulation complies with the physicochemical tests, as described in the current USP General Chapter [381] "Elastomeric Closures for Injections".
The Westar RS components receive a Water for Injection (WFI) final rinse.

This product does not contain any intentionally added lead, cadmium, hexavalent chromium, or mercury. In accordance with CONEG, West certifies that any incidental levels of these metals will not exceed 100 ppm in aggregate.
ELASTOMER TYPE: CHLOROBUTYL

The formulation used in this product is not made with latex or dry rubber.
This product does not contain 2-Mercaptobenzothiazole (2-MCBT) or any related substances.
West certifies that the formulation conforms to the formulation specifications.

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QUALITY CERTIFICATE

No. 80372784 - 000010

Batch / Lot Identification

S.No.	Batch/Lot Number	Manufacturing Date	Batch Specific Information
1	D000255363	05 MAR 2025	N/A
2	D000255367	05 MAR 2025	N/A
3	D000255372	05 MAR 2025	N/A
4	D000255378	05 MAR 2025	N/A
5	D000255381	05 MAR 2025	N/A
6	D000255386	05 MAR 2025	N/A

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QUALITY CERTIFICATE

No. 80372784 - 000010

Analysis Section

AA-29: B2 Level [ug/part]	>= 0
Bioburden [CFU / 100 cm2]	<= 5
Bacterial Endotoxins (<= 1 EU/piece)	Test PASS or FAIL
Endotoxins - For Info Only (EU/piece)	
Particl-08: >25-50 um Post Wash Calc [Particles/10cm2]	<= 13.0
Particl-08: >50-100 um Post Wash Calc [Particles/10cm2]	<= 3.5
Particl-08: >100 um Post Wash Calc [Particles/10cm2]	<= 0.9
Particl-08: PCI Post Wash Calc	<= 2.5

Actual Data

Test	Batch/Lot No	Batch/Lot No	Batch/Lot No
	D000255363	D000255367	D000255372
AA-29: B2 Level	108	104	106
Bioburden	0	0	0
Bacterial Endotoxins (<= 1 EU/piece)	PASS	PASS	PASS
Endotoxins - For Info Only (EU/piece)	< 0.05	< 0.06	< 0.05
Particl-08: >25-50 um Post Wash Calc	0.2	0.2	0.1
Particl-08: >50-100 um Post Wash Calc	0.1	0.0	0.1
Particl-08: >100 um Post Wash Calc	0.0	0.0	0.1
Particl-08: PCI Post Wash Calc	0.0	0.0	0.1

Test	Batch/Lot No	Batch/Lot No	Batch/Lot No
	D000255378	D000255381	D000255386
AA-29: B2 Level	115	117	117
Bioburden	0	0	0
Bacterial Endotoxins (<= 1 EU/piece)	PASS	PASS	PASS
Endotoxins - For Info Only (EU/piece)	< 0.05	< 0.05	< 0.05
Particl-08: >25-50 um Post Wash Calc	0.2	0.8	0.1

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QUALITY CERTIFICATE

No. 80372784 - 000010

Test	Batch/Lot No	Batch/Lot No	Batch/Lot No
	D000255378	D000255381	D000255386
Particl-08: >50-100 um Post Wash Calc	0.1	0.1	0.0
Particl-08: >100 um Post Wash Calc	0.0	0.0	0.0
Particl-08: PCI Post Wash Calc	0.0	0.1	0.0

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QUALITY CERTIFICATE

No. 80372784 - 000010

Other Information

Bioburden 01 [CFU / 100 cm2]	<= 5.000

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West Pharmaceutical Services, Inc.
530 Herman O. West Drive
Exton, PA 19341
www.westpharma.com

Compliance Bulletin

Rev.11

West Item: 4432/50 Grey

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West Pharmaceutical Services, Inc.
530 Herman O. West Drive
Exton, PA 19341
www.westpharma.com

West Item: 4432/50 Grey

1 INTRODUCTION

This Compliance Bulletin provides key information requested by customers on the West Pharmaceutical Services (West) Item herein. This information is offered in good faith and is not to be considered a warranty or quality specification. Because final conditions and use of this item are beyond our control, this information should not be substituted for customer's own compliance determinations and tests to assure this item is appropriate for any end use.

Compliance Bulletins are provided upon customer request. Please accept this in lieu of West completing questionnaires or surveys. If you have additional compliance questions as a result of an Agency request, please provide West's Regulatory Team an appropriately redacted copy of the Agency's request with reference to the applicable legislation so we can review it and provide a response.

This revision shall remain valid unless a change is made to a product or process impacting the statements herein. Change notifications, where required, are managed in accordance with West's change control process.

2 ANIMAL DERIVED MATERIALS

- **EMA/410/01 rev. 3, Note for guidance on minimising the risk of transmitting animal spongiform encephalopathy agents via human and veterinary medicinal products**

The ingredients used in the manufacture of this formulation have been evaluated for animal sourcing based upon raw material supplier declarations. No materials of animal origin are directly sourced by West as primary ingredients. Based upon supplier declarations, no secondary source materials are from animal origin.

West has determined that this formulation does not have any BSE/TSE risk since there are no materials of animal origin added; thus the requirements of EMA/410/01 rev. 3: *Note for guidance on minimizing the risk of transmitting animal spongiform encephalopathy agents via human and veterinary medicinal products* do not apply.

3 HEAVY METALS

- **EUROPEAN DIRECTIVE 94/62/EC DATED 20 DECEMBER 1994, ON PACKAGING AND PACKAGING WASTE**
- **THE MODEL TOXICS IN PACKAGING LEGISLATION (FORMERLY KNOWN AS CONEG)[1]**

The item does not contain intentionally added lead, cadmium, hexavalent chromium, or mercury. In addition, West certifies that any incidental levels of these metals will not exceed the following limit:

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Lead Cadmium Hexavalent Chromium Mercury	
as a sum	100 ppm maximum

[1]The Model Toxics in Packaging Legislation as detailed at: http://toxicsinpackaging.org/model-legislation/model/

4 NATURAL RUBBER LATEX/DRY NATURAL RUBBER CONTENT

The item **is not made** with Natural Rubber Latex or Dry Natural Rubber.

5 ADDITIVES OF CONCERN

The following additives of concern are not intentionally added to the item:

- Aflatoxins
- Asbestos
- Bisphenol A (BPA)
- Dioxins
- Gluten
- Glycerine or oils derived from the *Jatropha* plant
- Lactose
- Melamine
- 2-Mercaptobenzothiazole (2-MCBT)
- Nitrosamines
- Ozone Depleting Substances
- Parabens
- Persistent Organic Pollutants per Stockholm Convention
- Phthalates
- Polyvinylchlorides
- Thiuram Mix

The following additives of concern are present in the Item:

- Alcohol: One or more raw materials used in this item contains Alcohol.
- Butylated Hydroxytoluene (BHT): One or more raw materials used in this item contains BHT.
- Genetically Modified Organisms (GMOs): A raw material used in this item contains a substance which has been genetically modified. See below for West's position on GMOs for additional information
- Polycyclic Aromatic Hydrocarbons (PAHs): A raw material used in this item contains PAHs.

West's position on the following additives of concern:

Allergens and Excipients: There is currently no requirement to restrict or ban the use of these substances. Therefore, a general statement regarding Allergens cannot be generated. However, for the additives listed, West can state that these materials are not intentionally added to this West item.

Metal Catalyst: The EMEA/CHMP/SWP/4446/2000 (Limits for Residues of Metal Catalysts) is applicable to drug substance and excipients. In the scope of the guideline, metal catalysts are

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referred to as substances used in the synthesis of the drug substance or an excipient used in medicinal product. Therefore, West has not evaluated if this item with regard to metal catalysts.

Genotoxic Impurities: Agency and ICH guidelines on genotoxic impurities provides a general framework and practical approaches on dealing with genotoxic impurities in new active substances. For this reason, the guideline is not applicable to this item.

Residual Solvents: The focus of the ICH *Impurities: Guideline for Residual Solvents*, Q3C is on the solvents typically used for purification processes in the synthesis of active substances and in subsequent manufacturing steps (like e.g. granulation for tablets) of the finished dosage forms. This does not apply to the manufacturing of the West item. Therefore, West has not undertaken reviews on with regards to the residual solvents defined in the ICH Guideline and thus, cannot certify.

GMO Genetically Modified Organisms: The guideline EMEA/CHMP/BWP/135148/2004 (Environmental Risk Assessments for Medicinal Products Containing, or consisting of, Genetically Modified Organisms) and such related guidelines relate to medicinal products, specifically to the active ingredients and excipients a medicinal product is made. Therefore, the guideline is not applicable to this item.

6 ROHS

- **European Directive 2011/65/EU of 8 June 2011 on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS 2), as amended by 2015/863/EU**

Based upon information provided by our suppliers, West can certify that to our knowledge the ingredients in the aforementioned formulation are in compliance with the requirements presented in European Directive 2011/65/EU as amended by 2015/863/EU on the restriction of the use of certain hazardous substances in electrical and electronic equipment.

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7 FOOD REGULATIONS

Please note: This West Item is not intended to directly or in-directly contact food.

- **21 CFR Parts 174-186**

The Food and Drug Administration outlines regulations for Food Additives under 21 CFR 174 - 186. West Pharmaceutical Services (West) reviews raw materials against these regulations for pharmaceutical packaging components.

A formulation containing a material not listed in these sections does not preclude it from use; however, use of a substance included in the Food Additive section may expedite the process of obtaining pre-market approval.

Finally, one or more ingredients in this item can be found in the Food Additives section under 21 CFR 174-186.

- **Commission Regulation (EU) No 10/2011 of 14 January 2011 on plastic materials and articles intended to come into contact with food**

The European Food Regulation (EU) No 10/2011 of 14 January 2011 on plastic materials and articles intended to come into contact with food clearly excludes rubber items. Please refer to the following:

This Regulation shall not apply to the following materials and articles which are placed on the EU market and are intended to be covered by other specific measures:

(a) ion exchange resins;
*(b) **rubber**;*
(c) silicones.

Therefore, West has not undertaken reviews with regards to Commission Regulation (EU) No 10/2011 and thus, cannot certify.

8 PROPOSITION 65

- **Proposition 65, the Safe Drinking Water and Toxic Enforcement Act of 1986**

California's Proposition 65 is a list of chemicals known to the state to cause cancer or reproductive toxicity. A supplier of a raw material in this formula has confirmed the presence of the following substances listed on California's Proposition 65 List as dated 23 APRIL 2021.

Arsenic: trace
Beryllium: trace
Cadmium: trace

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Hexavalent chromium: trace
Lead: trace
Mercury: trace
Nickel: trace

9 ELEMENTAL IMPURITIES ICH Q3D: Guideline for Elemental Impurities, dated 16 Dec 2014

ICH Q3D guidelines apply to applicants of final drug product but recognize direct and/or indirect sources of metals from excipients, catalysts, from components of container closure systems and drug manufacturing equipment.

However, one or more suppliers have declared the following **Class 1, Class 2A, Class 2B, and Class 3** elements are present:

ICH Q3D Class	Element
1	Arsenic
1	Lead
2A	Cobalt
2A	Nickel
2A	Vanadium
2B	Silver
2B	Selenium
3	Chromium
3	Copper
3	Lithium
3	Molybdenum
3	Antimony

Due to the nature of some used raw materials, this does not guarantee that trace amounts of impurities will not be present. Additionally, one or more raw material suppliers have not provided feedback for every element.

West has generated Extractable Elements Data Sheets for some items. Please contact **West Technical Support** for additional information on our Extractable Elements Data Sheets.

10 HALAL

This item is not certified as Halal.

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530 Herman O. West Drive
Exton, PA 19341
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Compliance Bulletin

Rev.11

West Item: 4432/50 Grey

11 KOSHER

This item is not certified as Kosher.

Cathy Chhour
Regulatory Affairs Manager, Product Stewardship
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cathy.chhour@westpharma.com

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Compliance Bulletin

Rev. 2

West Item: FluroTec Resin D

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530 Herman O. West Drive
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West Item: FluroTec Resin D

1 INTRODUCTION

This Compliance Bulletin provides key information requested by customers on the West Pharmaceutical Services (West) Item herein. This information is offered in good faith and is not to be considered a warranty or quality specification. Because final conditions and use of this item are beyond our control, this information should not be substituted for customer's own compliance determinations and tests to assure this item is appropriate for any end use.

Compliance Bulletins are provided upon customer request. Please accept this in lieu of West completing questionnaires or surveys. If you have additional compliance questions as a result of an Agency request, please provide West's Regulatory Team an appropriately redacted copy of the Agency's request with reference to the applicable legislation for review and a response will be provided.

This revision shall remain valid unless a change is made to a product or process impacting the statements herein. Change notifications, where required, are managed in accordance with West's change control process.

2 ANIMAL DERIVED MATERIALS

- **EMA/410/01 rev. 3, *Note for guidance on minimising the risk of transmitting animal spongiform encephalopathy agents via human and veterinary medicinal products***

The ingredients used in the manufacture of this film have been evaluated for animal sourcing based upon raw material supplier declarations. No materials of animal origin are directly sourced by West as primary ingredients. Based upon supplier declarations, no secondary source materials are from animal origin.

West has determined that this film does not have any BSE/TSE risk since there are no materials of animal origin added; thus the requirements of EMA/410/01 rev. 3: Note for guidance on minimizing the risk of transmitting animal spongiform encephalopathy agents via human and veterinary medicinal products do not apply.

3 HEAVY METALS

- **EUROPEAN DIRECTIVE 94/62/EC DATED 20 DECEMBER 1994, ON PACKAGING AND PACKAGING WASTE**
- **THE MODEL TOXICS IN PACKAGING LEGISLATION (FORMERLY KNOWN AS CONEG)[1]**

The item does not contain intentionally added lead, cadmium, hexavalent chromium, or mercury. In addition, West certifies that any incidental levels of these metals will not exceed the following limit:

Lead

Cadmium

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Hexavalent Chromium Mercury	as a sum 100 ppm maximum

[1]The Model Toxics in Packaging Legislation as detailed at: http://toxicsinpackaging.org/model-legislation/model/

4 NATURAL RUBBER LATEX/DRY NATURAL RUBBER CONTENT

The item **is not made** with Natural Rubber Latex or Dry Natural Rubber.

5 ADDITIVES OF CONCERN

The following additives of concern are not intentionally added to the item:

- Aflatoxins
- Alcohol
- Asbestos
- Bisphenol A (BPA)
- Butylated Hydroxytoluene (BHT)
- Dioxins
- Genetically Modified Organisms
- Gluten
- Glycerine or oils derived from the Jatropha plant
- Lactose
- Melamine
- 2-Mercaptobenzothiazole (2-MCBT)
- Nitrosamines
- Ozone Depleting Substances
- Parabens
- Persistent Organic Pollutants per Stockholm Convention
- Phthalates
- Polynuclear Aromatic Hydrocarbons (PAHs or PNAs)
- Polyvinylchlorides
- Thiuram

West's position on the following additives of concern:

- **Allergens and Excipients:** There is currently no requirement to restrict or ban the use of these substances. Therefore, a general statement regarding Allergens cannot be generated. However, for the additive listed, West can state that these materials are not intentionally added to this West item.
- **Metal Catalyst:** The EMEA/CHMP/SWP/4446/2000 (Limits for Residues of Metal Catalysts) is applicable to drug substance and excipients. In the scope of the guideline, metal catalysts are referred to as substances used in the synthesis of the drug substance or an excipient used in medicinal product. Therefore, West has not evaluated this item regarding metal catalysts.
- **Genotoxic Impurities:** Agency and ICH guidelines on genotoxic impurities provides a general framework and practical approaches on dealing with genotoxic impurities in new active substances. For this reason, the guideline is not applicable to this item.
- **Residual Solvents:** The focus of the ICH *Impurities: Guideline for Residual Solvents*, Q3C is on the solvents typically used for purification processes in the synthesis of active substances and in subsequent manufacturing steps (like e.g. granulation for tablets) of the finished dosage forms. This does not apply to the manufacturing of the West item.

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Therefore, West has not undertaken reviews on with regards to the residual solvents defined in the ICH Guideline and thus, cannot certify.

- **GMO Genetically Modified Organisms:** The guideline EMEA/CHMP/BWP/135148/2004 (Environmental Risk Assessments for Medicinal Products Containing, or consisting of, Genetically Modified Organisms) and such related guidelines relate to medicinal products, specifically to the active ingredients and excipients a medicinal product is made. Therefore, the guideline is not applicable to this item.

6 ROHS

- **European Directive 2011/65/EU of 8 June 2011 on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS 2), as amended by 2015/863/EU**

Based upon information provided by our suppliers, West can certify that to our knowledge the ingredients in the aforementioned film are in compliance with the requirements presented in European Directive 2011/65/EU as amended by 2015/863/EU on the restriction of the use of certain hazardous substances in electrical and electronic equipment.

7 FOOD REGULATIONS

Please note: This West Item is not intended to directly or in-directly contact food.

- **Commission Regulation (EU) No 10/2011 of 14 January 2011 on plastic materials and articles intended to come into contact with food**

The European Food Regulation (EU) No 10/2011 of 14 January 2011 on plastic materials and articles intended to come into contact with food clearly excludes rubber items. Please refer to the following:

This Regulation shall not apply to the following materials and articles which are placed on the EU market and are intended to be covered by other specific measures:

(a) ion exchange resins;
*(b) **rubber**;*
(c) silicones.

Therefore, West has not undertaken reviews with regards to Commission Regulation (EU) No 10/2011 and thus, cannot certify.

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West Pharmaceutical Services, Inc.
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8 PROPOSITION 65

- **Proposition 65, the Safe Drinking Water and Toxic Enforcement Act of 1986**

California's Proposition 65 is a list of chemicals known to the state to cause cancer or reproductive toxicity. This item does not contain chemicals on California's Proposition 65 List as dated 23 Dec 2021.

9 ELEMENTAL IMPURITIES

- **ICH Q3D: Guideline for Elemental Impurities, dated 16 Dec 2014**

ICH Q3D guidelines apply to applicants of final drug product but recognize direct and/or indirect sources of metals from excipients, catalysts, from components of container closure systems and drug manufacturing equipment.

West **does not** intentionally add **Class 1, Class 2A, Class 2B or Class 3** elements as defined by ICH Q3D to the item listed herein.

Due to the nature of some used raw materials, this does not guarantee that trace amounts of impurities will not be present.

West does not monitor extractable elements analytically for this item and, therefore cannot provide quantitative data. West Analytical Services can support risk assessments for elemental impurities in drug products (leachables) as well as potential for extractable elements for intended use (extractable simulation) to correlate to final drug product. Please contact **West Technical Support** and/or **West Analytical Services** for additional information.

10 HALAL

This item is not certified as Halal.

11 KOSHER

This item is not certified as Kosher.

Jennifer Perks
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Jennifer.Perks@westpharma.com

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Compliance Bulletin

Rev. 5

West Item: B2 Coating

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West Pharmaceutical Services, Inc.
530 Herman O. West Drive
Exton, PA 19341
www.westpharma.com

West Item: B2 Coating

1 INTRODUCTION

This Compliance Bulletin provides key information requested by customers on the West Pharmaceutical Services (West) Item herein. This information is offered in good faith and is not to be considered a warranty or quality specification. Because final conditions and use of this item are beyond our control, this information should not be substituted for customer's own compliance determinations and tests to assure this item is appropriate for any end use.

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This revision shall remain valid unless a change is made to a product or process impacting the statements herein. Change notifications, where required, are managed in accordance with West's change control process.

2 ANIMAL DERIVED MATERIALS

- **EMA/410/01 rev. 3, *Note for guidance on minimising the risk of transmitting animal spongiform encephalopathy agents via human and veterinary medicinal products***

The ingredients used in the manufacture of this coating have been evaluated for animal sourcing based upon raw material supplier declarations. No materials of animal origin are directly sourced by West as primary ingredients. Based upon supplier declarations, no secondary source materials are from animal origin.

West has determined that this coating does not have any BSE/TSE risk since there are no materials of animal origin added; thus the requirements of EMA/410/01 rev. 3: *Note for guidance on minimizing the risk of transmitting animal spongiform encephalopathy agents via human and veterinary medicinal products* do not apply.

3 HEAVY METALS

- **EUROPEAN DIRECTIVE 94/62/EC DATED 20 DECEMBER 1994, ON PACKAGING AND PACKAGING WASTE**
- **THE MODEL TOXICS IN PACKAGING LEGISLATION (FORMERLY KNOWN AS CONEG)[1]**

The item does not contain intentionally added lead, cadmium, hexavalent chromium, or mercury. In addition, West certifies that any incidental levels of these metals will not exceed the following limit:

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530 Herman O. West Drive
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Lead	
Cadmium	
Hexavalent Chromium	
Mercury	
as a sum	**100 ppm maximum**

[1]The Model Toxics in Packaging Legislation as detailed at: http://toxicsinpackaging.org/model-legislation/model/

4 NATURAL RUBBER LATEX/DRY NATURAL RUBBER CONTENT

The item **is not made** with Natural Rubber Latex or Dry Natural Rubber

5 ADDITIVES OF CONCERN

The following additives of concern are not intentionally added to the item:

- Aflatoxins
- Asbestos
- Bisphenol A (BPA)
- Butylated Hydroxytoluene (BHT)
- Dioxins
- Genetically Modified Organisms
- Gluten
- Glycerine or oils derived from the Jatropha plant
- Lactose
- Melamine
- 2-Mercaptobenzothiazole (2-MCBT)
- Nitrosamines
- Ozone Depleting Substances
- Parabens
- Persistent Organic Pollutants per Stockholm Convention
- Phthalates
- Polycyclic Aromatic Hydrocarbons (PAH)
- Polyvinylchlorides
- Thiuram

West's position on the following additives of concern:

- **Allergens and Excipients:** There is currently no requirement to restrict or ban the use of these substances. Therefore, a general statement regarding Allergens cannot be generated. However, for the additives listed, West can state that these materials are not intentionally added to this West item
- **Metal Catalyst:** The EMEA/CHMP/SWP/4446/2000 (Limits for Residues of Metal Catalysts) is applicable to drug substance and excipients. In the scope of the **guideline,** metal catalysts are referred to as substances used in the synthesis of the drug substance or an excipient used in medicinal product. Therefore, West has not evaluated this itemregarding metal catalysts.
- **Genotoxic Impurities:** Agency and ICH guidelines on genotoxic impurities provides a general framework and practical approaches on dealing with genotoxic impurities in new active substances. For this reason, the guideline is not applicable to this item.
- **Residual Solvents:** The focus of the ICH *Impurities: Guideline for Residual Solvents*, Q3C is on the solvents typically used for purification processes in the synthesis of active

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substances and in subsequent manufacturing steps (like e.g. granulation for tablets) of the finished dosage forms. This does not apply to the manufacturing of the West item. Therefore, West has not undertaken reviews on with regards to the residual solvents defined in the ICH Guideline and thus, cannot certify.

- **GMO Genetically Modified Organisms:** The guideline EMEA/CHMP/BWP/135148/2004 (Environmental Risk Assessments for Medicinal Products Containing, or consisting of, Genetically Modified Organisms) and such related guidelines relate to medicinal products, specifically to the active ingredients and excipients a medicinal product is made. Therefore, the guideline is not applicable to this item.

6 ROHS

- **European Directive 2011/65/EU of 8 June 2011 on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS 2), as amended by 2015/863/EU**

Based upon information provided by our suppliers, West can certify that to our knowledge the ingredients in the aforementioned coating are in compliance with the requirements presented in European Directive 2011/65/EU as amended by 2015/863/EU on the restriction of the use of certain hazardous substances in electrical and electronic equipment.

7 FOOD REGULATIONS

Please note: This West Item is not intended to directly or in-directly contact food.

- **21 CFR Parts 174-186**

The Food and Drug Administration outlines regulations for Food Additives under 21 CFR 174 - 186. West Pharmaceutical Services (West) reviews raw materials against these regulations for pharmaceutical packaging components.

Material not listed in these sections does not preclude it from use; however, use of a substance included in the Food Additive section may expedite the process of obtaining pre-market approval.

Finally, one or more ingredients in this item can be found in the Food Additives section under 21 CFR 174-186.

- **Commission Regulation (EU) No 10/2011 of 14 January 2011 on plastic materials and articles intended to come into contact with food**

The European Food Regulation (EU) No 10/2011 of 14 January 2011 on plastic materials and articles intended to come into contact with food clearly excludes rubber items. Please refer to the following:

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This Regulation shall not apply to the following materials and articles which are placed on the EU market and are intended to be covered by other specific measures:

(a) ion exchange resins;
*(b) **rubber**;*
(c) silicones.

Therefore, West has not undertaken reviews with regards to Commission Regulation (EU) No 10/2011 and thus, cannot certify.

8 PROPOSITION 65

- **Proposition 65, the Safe Drinking Water and Toxic Enforcement Act of 1986**

California's Proposition 65 is a list of chemicals known to the state to cause cancer or reproductive toxicity. This item does not contain chemicals on California's Proposition 65 List as dated 15 March 2021.

9 ELEMENTAL IMPURITIES

- **ICH Q3D: Guideline for Elemental Impurities, dated 16 Dec 2014**

ICH Q3D guidelines apply to applicants of final drug product but recognize direct and/or indirect sources of metals from excipients, catalysts, from components of container closure systems and drug manufacturing equipment.

Class 1, Class 2A, Class 2B or Class 3 elements as defined by ICH Q3D **are not** intentionally added to the item listed herein.

Due to the nature of some used raw materials, this does not guarantee that trace amounts of impurities will not be present.

West does not monitor extractable elements analytically for this item and, therefore cannot provide quantitative data. West Analytical Services can support risk assessments for elemental impurities in drug products (leachables) as well as potential for extractable elements for intended use (extractable simulation) to correlate to final drug product. Please contact **West Technical Support** and/or **West Analytical Services** for additional information.

10 HALAL

This item is not certified as Halal.

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West Pharmaceutical Services, Inc.
530 Herman O. West Drive
Exton, PA 19341
www.westpharma.com

West Item: B2 Coating

11 KOSHER

This item is not certified as Kosher.

Cathy Chhour
Regulatory Affairs Manager, Product Stewardship
M: +1 610-850-2743 F: +1 610-717-0668
cathy.chhour@westpharma.com

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Cathy Chhour
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Overview of FluroTec® and B2-Coating

I. Background

Through West Pharmaceutical Services, (West) partnership with Daikyo Seiko, Ltd. (Daikyo), elastomeric closures are offered by West with FluroTec barrier film and B2-coating. Closures may have only FluroTec film or only B2-coating applied; or may have both FluroTec film and B2-coating applied to different closure surface areas. FluroTec and B2-coated products are widely used by the global pharmaceutical industry. The FluroTec film provides a barrier to leachables and extractables, thus mitigating the risk of interaction over the drug product shelf life. The FluroTec film, in combination with the B2-coating, provides lubricity without the need for free silicone oil, eliminating a potential source of particulate contamination.

II. FluroTec Film

FluroTec is a transparent fluoropolymer film (ETFE, ethylene tetrafluoroethylene) that is applied during the molding process. It is conformable to complex shaped closures and pistons, and remains intact during normal closure processing conditions. The film can be applied to the top and bottom surfaces of a closure or piston. The FluroTec film on the bottom surface (plug portion) of the closure provides an effective barrier against organic and inorganic extractables, minimizing interaction between the drug and the closure while maintaining container closure integrity. The film reduces absorption and adsorption of the drug product, an important benefit for maintaining the strength and shelf life of most drugs. FluroTec film on the top surface of the closure helps prevent the closures from sticking to lyophilization chamber shelves and also reduces closure clumping in sterilizers and feeder bowls. This, in turn, reduces product losses caused by closure sticking and keeps production on schedule.

Here is a depiction of a closure with FluroTec film applied to the top and bottom surfaces. The dark gray color indicates the area of FluroTec coverage and is for illustration purposes only.

III. B2-Coating

B2-Coating is a combination of high molecular weight silicones (cross-linkable polydimethylsiloxane) applied by spray, which is then heated and UV cured to the surface of elastomeric closures. In comparison to free silicone oil, the B2-Coating greatly reduces particulate levels while enhancing closure lubricity.

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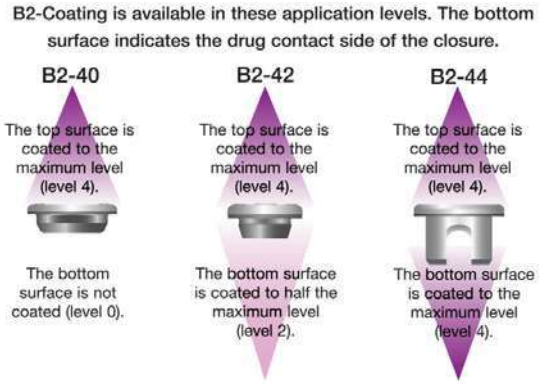
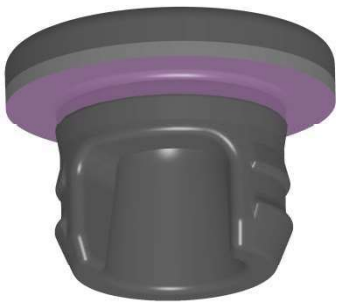
Overview of FluroTec® and B2-Coating

Many levels of B2-coating can be applied to the top and bottom surfaces of the closure for optimization of lubricity and machinability. The Level 0 designates no B2-coating, Level 2 is a medium application and Level 4 is the maximum application level. The following is a depiction of the levels of application.



An additional, and widely used, B2-coating application offered by West is B2-TR. B2-TR is most commonly offered for lyophilization closures, which have FluroTec film applied to the top and bottom surfaces of the closure. The B2-coating is applied in a trace amount to the area of the closure right under the flange (top) and right at the top of the plug portion of the closure (the closure/vial sealing area). The B2-coating is applied to this area to help ease the seating of the closure into the vial during the lyophilization process. Below is a depiction of the B2-TR application on a closure (with FluroTec film on the top and plug portion). The dark gray color indicates the area of FluroTec film coverage; the purple color indicates the area of B2-TR coverage and is for illustration purposes only.



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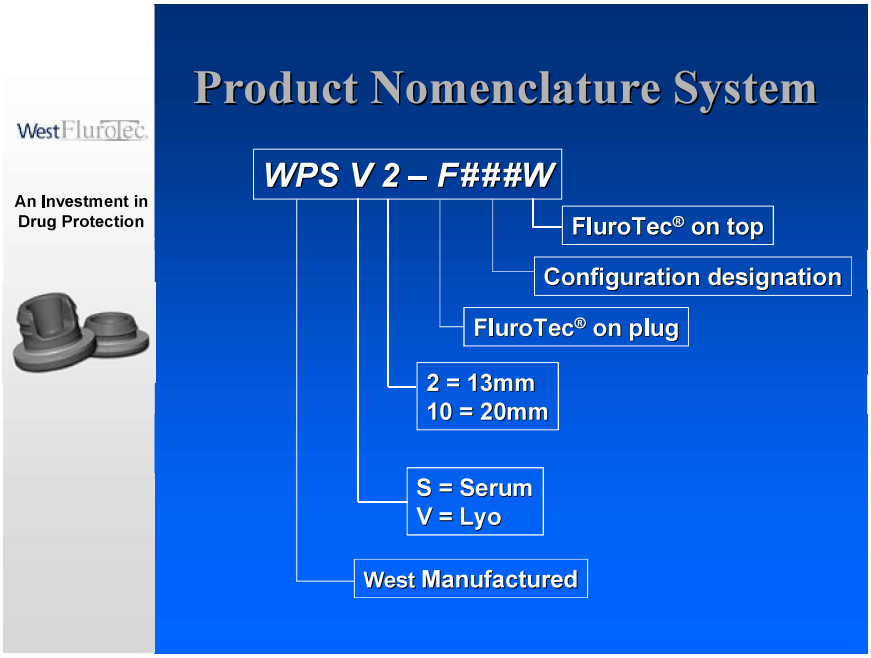
IV. FluroTec Product Nomenclature

The product descriptions for the West FluroTec closures originated from Daikyo product descriptions. Once explained, the product description designations are easily understood. Below is the explanation of the FluroTec product nomenclature.



FluroTec film and B2-coating meet their respective applicable pharmacopoeial requirements of the relevant global regulatory agencies.

West's products are sold on the basis that it is the customer's responsibility to evaluate and test the West product to determine its compatibility with other materials and fitness for any end use.

This technical bulletin dated 23 November 2009, supersedes any other previously released versions of this bulletin.

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FluroTec and B2-Coating technologies are licensed from Daikyo Seiko, Ltd.

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Version: Revision 3
Supersedes: Revision 2

FORMULATION CHARACTERISTICS

WEST FORMULATION 4432/50 GRAY

*Note: The formulations listed above conform to the specifications and properties represented below as an example.

1. FORMULATION SPECIFICATIONS

The following tests and specifications may be used to confirm the composition of the base elastomeric formulation. The ash and specific gravity specifications listed do not apply to closures with Teflon®, FluroTec® or LyoTec™ coatings.

Test	Units	Specification	Method
Ash	%	46.6 ± 2.0	West Ash-01
Specific Gravity	None	1.33 ± 0.04	West SpGrav-02
Ultraviolet (UV) Spectrum of NaOH Extract	None	Compares to Reference (Attachment 1)	West UV-01
Infrared (IR) Spectrum of Pyrolysis Products	None	Compares to Reference (Attachment 2)	West IR-01

The identity of coatings present may be verified using Reflectance Infrared analysis.

Spectrum	Attachment #	Method
Reflectance Infrared Analysis of Base Formulation	Attachment 3	West IR-08
Reflectance Infrared Analysis of Flurotec Surface	Attachment 4	West IR-08
Reflectance Infrared Analysis of B2 Surface	Attachment 5	West IR-08
Reflectance Infrared Analysis of Teflon 2 Surface	Attachment 6	West IR-08
Reflectance Infrared Analysis of Silicone Surface	Attachment 7	West IR-08
Reflectance Infrared Analysis of LyoTec Surface	Attachment 8	West IR-08
Reflectance Infrared Analysis of Base Surface	Attachment 9	West IR-26
Reflectance Infrared Analysis of FluroTec Surface	Attachment 10	West IR-26

2. GENERAL DESCRIPTION

The following is a general description of the composition of the base formulation.

Elastomer Type:	Chlorobutyl
Reinforcement System:	Inert Mineral
Curing System:	Unconventional
Dry Natural Rubber Content:	0.0%

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FluroTec® and TrakStar® are registered trademarks of West Pharmaceutical Services, Inc. in the U.S. and other jurisdictions. FluroTec® technology is licensed from Daikyo Seiko, Ltd.
LyoTec is a trademark of West Pharmaceutical Services, Inc. in the U.S. and other jurisdictions

3. TYPICAL HARDNESS

Durometer measures the ability of an elastomeric formula to resist indentation under conditions that do not puncture the rubber. Durometer value shown here is typical of base elastomeric formulation. This data is based on defined test specimens and cannot be generated on closure configurations.

TABLE 1
TYPICAL HARDNESS

Test	Units	Test Specimen Type	Typical Data	Method
Durometer	Shore A Units	ASTM Test Plug	50	ASTM Section 9, D 2240

4. COMPLIANCE INFORMATION

- This formulation meets the requirements for Type I Closures specified in General Chapter <381> of the United States Pharmacopeia, *Elastomeric Closures for Injection,* and *General* Chapter 3.2.9 of the European Pharmacopoeia, *Rubber Closures for Containers for Aqueous Parenteral Preparations, for Powders and for Freeze-Dried Powders*. Typical Physicochemical and Biological data is presented in Tables 2 and 4.
- This formulation meets the requirements specified in General Tests Section 7.03 of the Japanese Pharmacopoeia, *Test for Rubber Closure for Aqueous Infusions.* Typical data is presented in Table 3.
- Regulatory compliance for CONEG Toxic Packaging legislation and 2-Mercaptobenzothiazole (2-MCBT) or any related substances can be found in our Regulatory Compliance Bulletins.

TYPICAL EXTRACTION CHARACTERISTICS

The following tests are specified in General Chapter <381> of the United States Pharmacopeia, and in General Chapter 3.2.9 of the European Pharmacopoeia, for the determination of the biological reactivity and extraction characteristics of elastomeric closures for containers for preparations for parenteral use. The functional tests included in the USP and Ph. Eur. chapters are sample configuration dependent and are therefore outside of the scope of this document. The data presented below are typical for this formulation, however results may differ due to normal process variations. Subsequent revisions of the compendia will be reviewed when published, however this document will be revised to reference the new edition numbers only if significant changes are made to the applicable chapters.

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FORMULA CHARACTERISTICS – 4432/50 GRAY (continued)

TABLE 2
TYPICAL PHYSICOCHEMICAL DATA

Test	Specifications Type I	Typical Data									
		Base Formula	With Teflon 2	With B2	With Flurotec	With LyoTec	Westar RS	Westar RU Steam	Westar RU Steam w/ B2, Flurotec	Westar RU Gamma	Westar RU Gamma w/ B2, Flurotec
Appearance, Opalescence	Sample solution is not more opalescent than Reference Suspension II (6 NTU)	0	0	0	0	0	0	0	0	0	0
Appearance, Color	Sample solution is not more intensely colored than Reference Solution GY_5	<= GY_5	<= GY_5	<= GY_5	<= GY_5	<= GY_5	<= GY_5	<= GY_5	<= GY_5	<= GY_5	<= GY_5
Acidity or Alkalinity	NMT 0.3mL 0.01M NaOH NMT 0.8mL 0.01M HCl	0.1 NaOH	0.1 NaOH	0.0	0.1 NaOH	0.1 NaOH	0.1 NaOH	0.1 NaOH	0.1 NaOH	0.1 NaOH	0.1 NaOH
Absorbance	NMT 0.2 Absorbance between 220 and 360nm	0.0	0.0	0.0	0.0	0.0	0.0	0.0.	0.0	0.0	0.0
Reducing Substances	NMT 3.0 mL 0.01N Potassium Permanganate	0.0	0.0	0.0	0.2	0.0	0.1	0.1	0.1	0.0	0.0
Ammonium	NMT 2 ppm	<=2	<=2	<=2	<=2	<=2	<=2	<=2	<=2	<=2	<=2
Extractable Zinc	NMT 5 µg/mL	ND	0	ND	ND	ND	ND	0	ND	0	0
Extractable Heavy Metals	NMT 2 ppm	<=2	<=2	<=2	<=2	<=2	<=2	<=2	<=2	<=2	<=2
Residue on Evaporation	NMT 2.0 mg	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.0	0.0
Volatile Sulfides	Black stain on paper is not more intense than that of standard	<= Std	<= Std	<= Std	<= Std	<= Std	<= Std	<= Std	<= Std	<= Std	<= Std

NMT = Not more than NLT = Not less than ND = none detected

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FORMULA CHARACTERISTICS 4432/50 GRAY (continued)

TYPICAL JP EXTRACTION CHARACTERISTICS

The following tests are specified in General Tests Section 7.03 of the Japanese Pharmacopoeia for the determination of the extraction characteristics of rubber closures for aqueous infusions. The data presented is typical for this formulation, however results may differ due to normal process variations.

TABLE 3
TYPICAL JP DATA

Test	Specification	Typical Data
Cadmium	Absorbance of sample solution is not more than that of standard solution (5ppm)	<= 5
Lead	Absorbance of sample solution is not more than that of standard solution (5ppm)	<= 5
Transparency at 430nm	NLT 99.0 %T	100.0
Transparency at 650nm	NLT 99.0 %T	100.0
pH	Difference between pH of blank and sample solution is NMT 1.0	0.1
Zinc	Absorbance of sample solution is not more than that of standard solution (0.5ppm)	Less than or equal to standard
Potassium Permanganate Reducing Substances	Difference in mL of 0.002M potassium permanganate required for the blank and sample solution is NMT 2.0	0.0
Residue on Evaporation	NMT 2.0 mg	0.0
UV Spectrum	Maximum absorbance between 220 and 350nm is NMT 0.20	0.00
Cytotoxicity	$IC_{50}(\%) \geq 90\%$	$IC_{50} > 90$

****Base** formulation meets the requirements specified in General Tests Section 7.03 of the Japanese Pharmacopoeia, *Test for Rubber Closure for Aqueous Infusions.*

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FORMULA CHARACTERISTICS 4432/50 GRAY (continued)

TABLE 4
TYPICAL BIOLOGICAL REACTIVITY DATA

Test	Specification	Units	Typical Data		Method
			Base Formula	w/ B2	
Biological Reactivity, In Vitro, MEM Elution Test	Response to Sample Preparation is not greater than Grade 2 (mildly reactive)	None	Meets USP <87> compliance Result: Grade 0	Meets USP <87> compliance Result: Grade 0	USP <87>

*The ISO10993-5 Elution Method and the USP <87> Elution Test are equivalent; triplicate cell monolayers are treated with sample eluate in the ISO method whereas duplicate monolayers are treated in the USP method.

**Only materials that fail the USP <87>, Biological Reactivity Tests, In Vitro specifications are candidates for testing of the USP <88>, Biological Reactivity Tests, In Vivo.

ATTACHMENTS:

Attachment 1: Ultraviolet reference spectrum of sodium hydroxide extract (UV-01) [1 page]
Attachment 2: Infrared reference spectrum of pyrolysis product (IR-01) [1 page]
Attachment 3: Reflectance Infrared reference spectrum of Base Formula (IR-08) [1 page]
Attachment 4: Reflectance Infrared reference spectrum of FluroTec Surface (IR-08) [1 page]
Attachment 5: Reflectance Infrared reference spectrum of B2 Surface (IR-08) [1 page]
Attachment 6: Reflectance Infrared reference spectrum of Teflon 2 Surface (IR-08) [1 page]
Attachment 7: Reflectance Infrared reference spectrum of Silicone Surface (IR-08) [1 page]
Attachment 8: Reflectance Infrared reference spectrum of LyoTec Surface (IR-08) [1 page]
Attachment 9: Reflectance Infrared reference spectrum of Base surface (IR-26) [1 page]
Attachment 10: Reflectance Infrared reference spectrum of FluroTec Surface (IR-26) [1 page]

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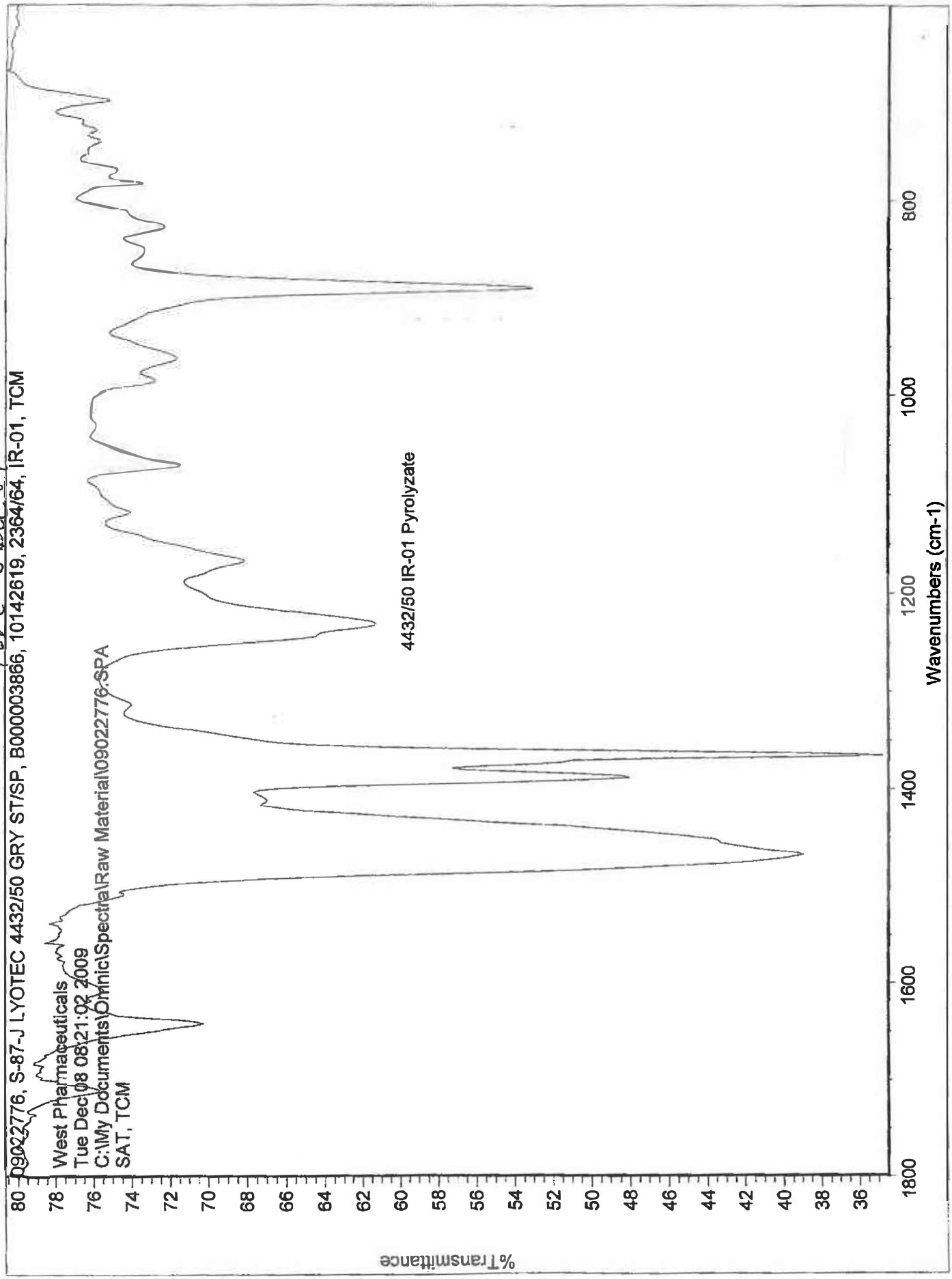

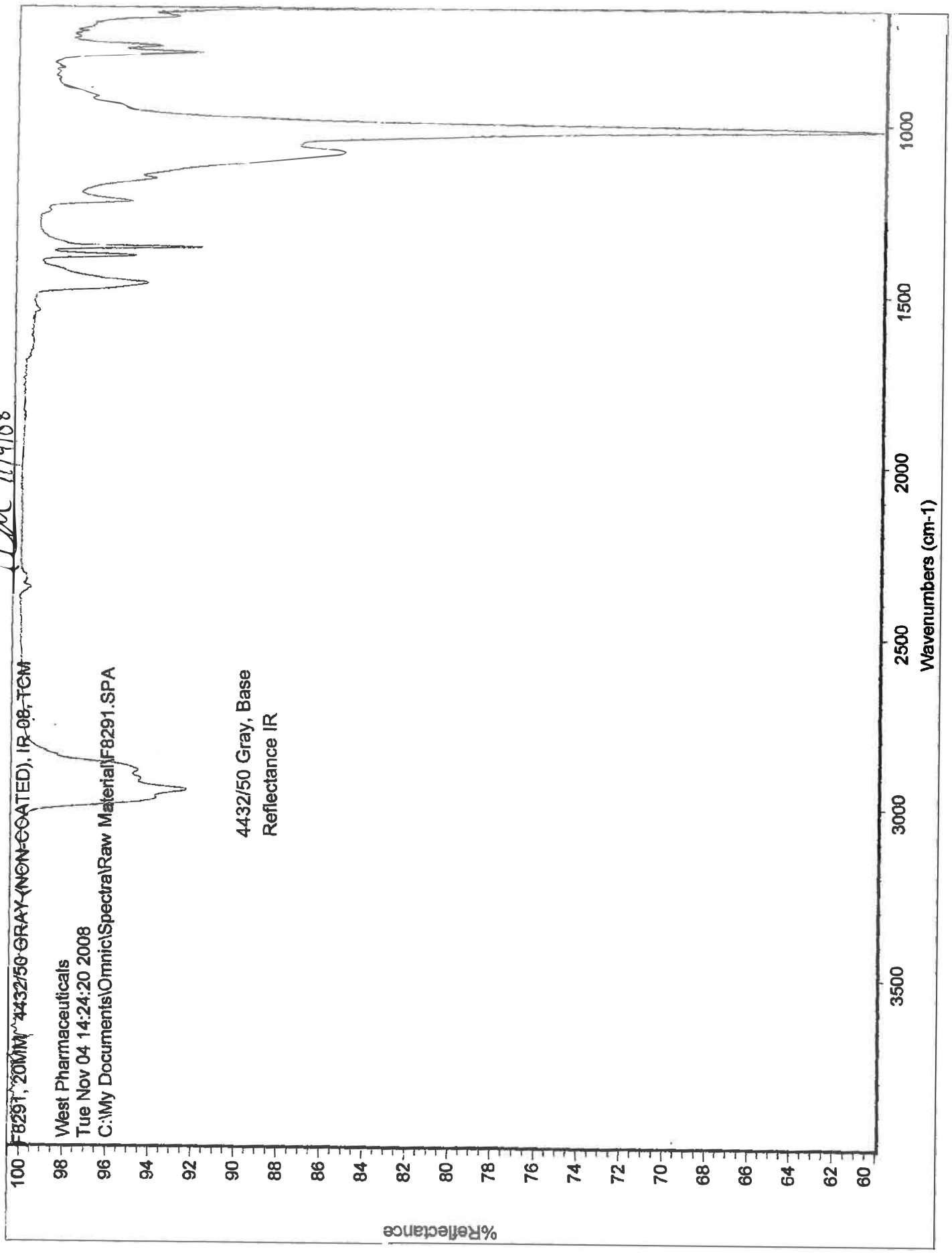

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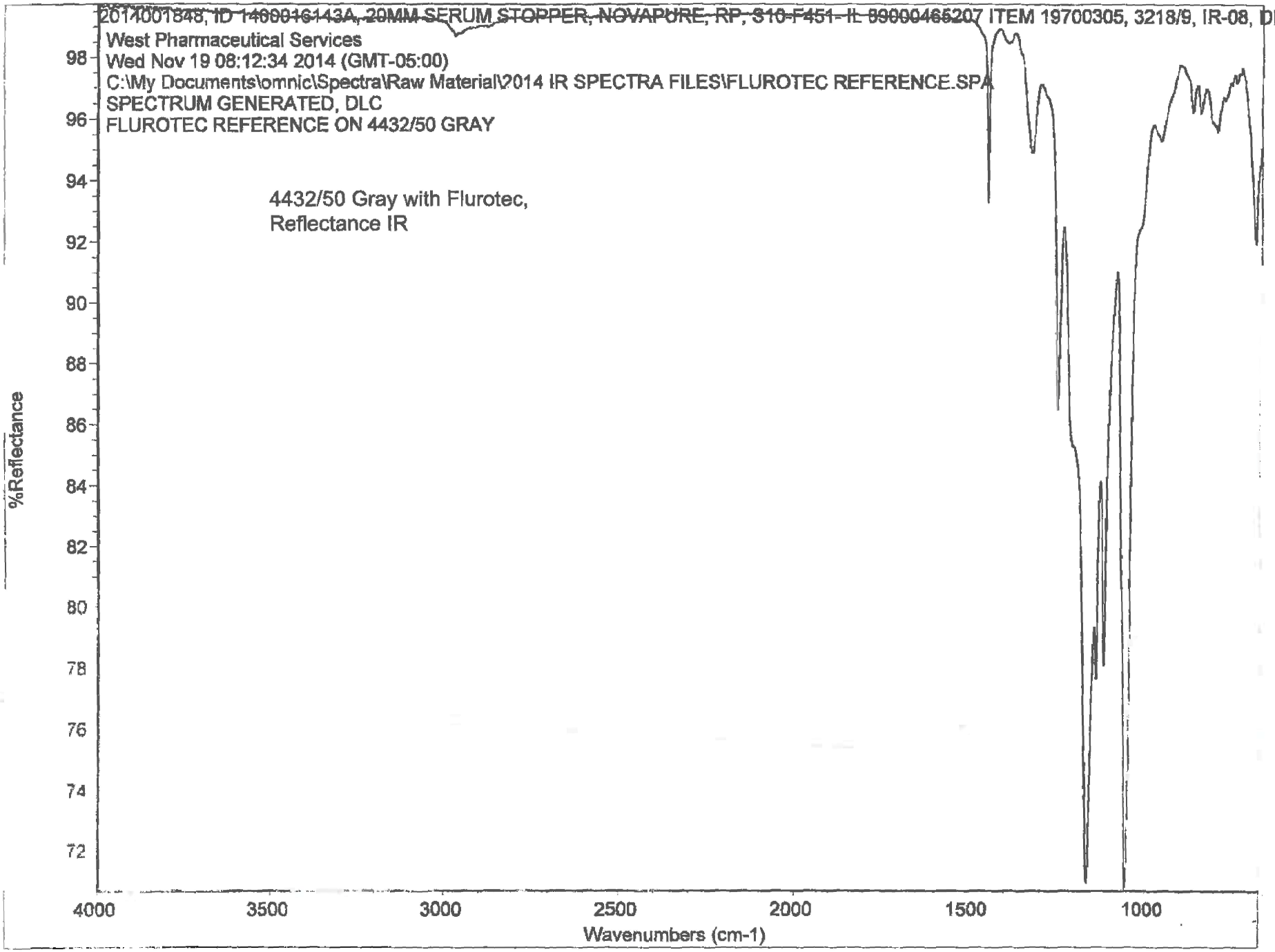
DLC 19 Nov 2014



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4432/50 Gray FC
Attachment 4

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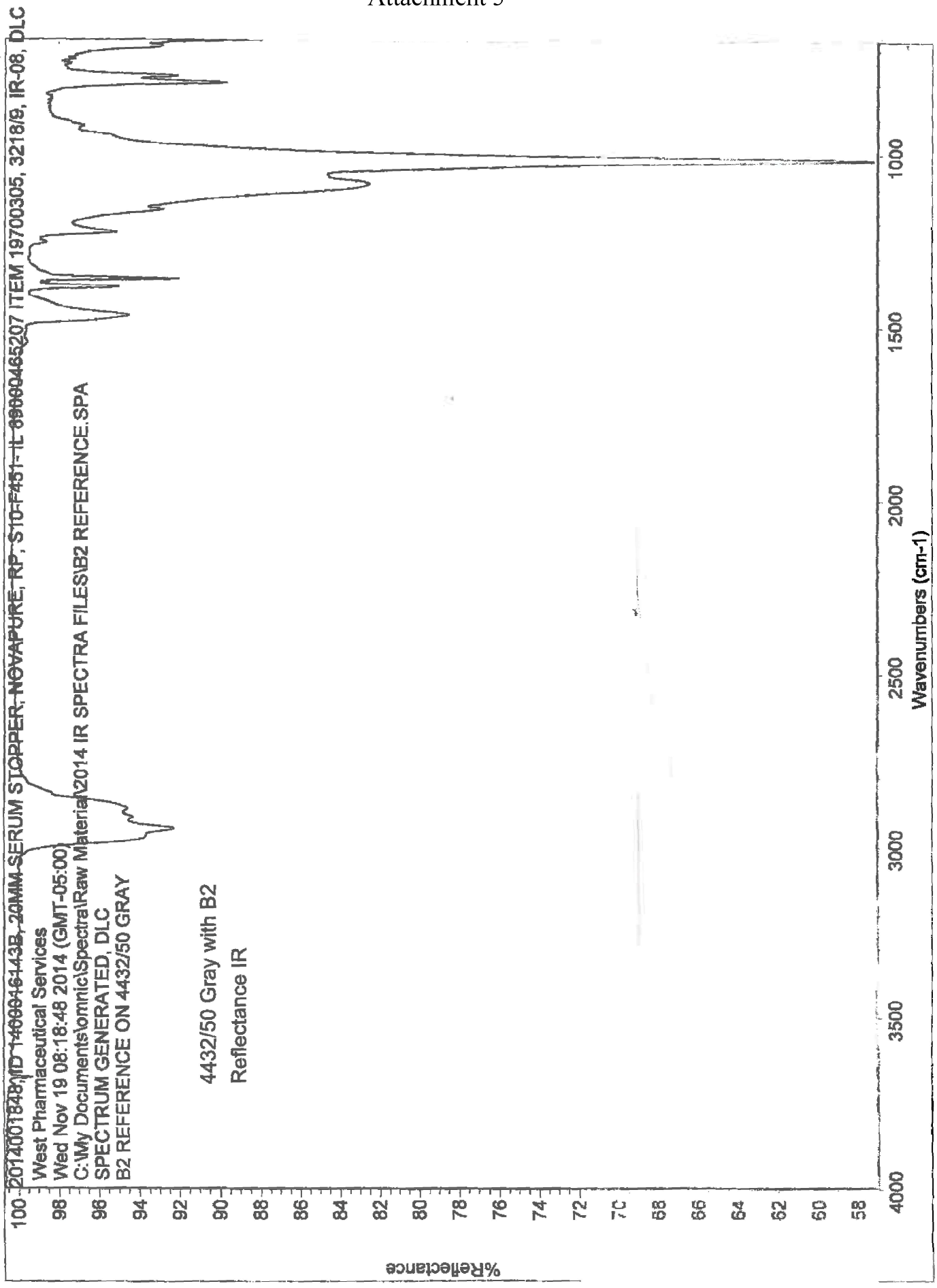

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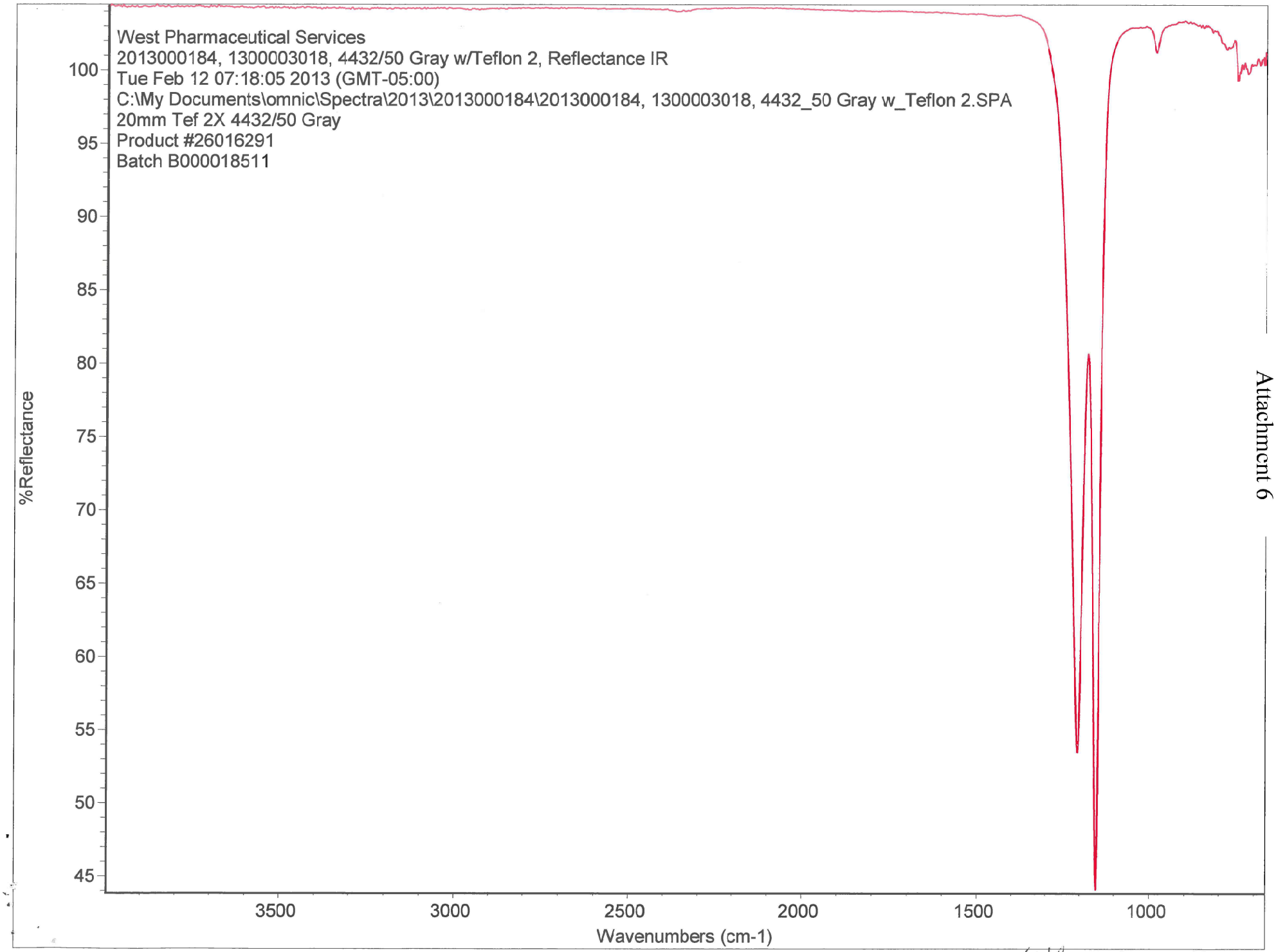
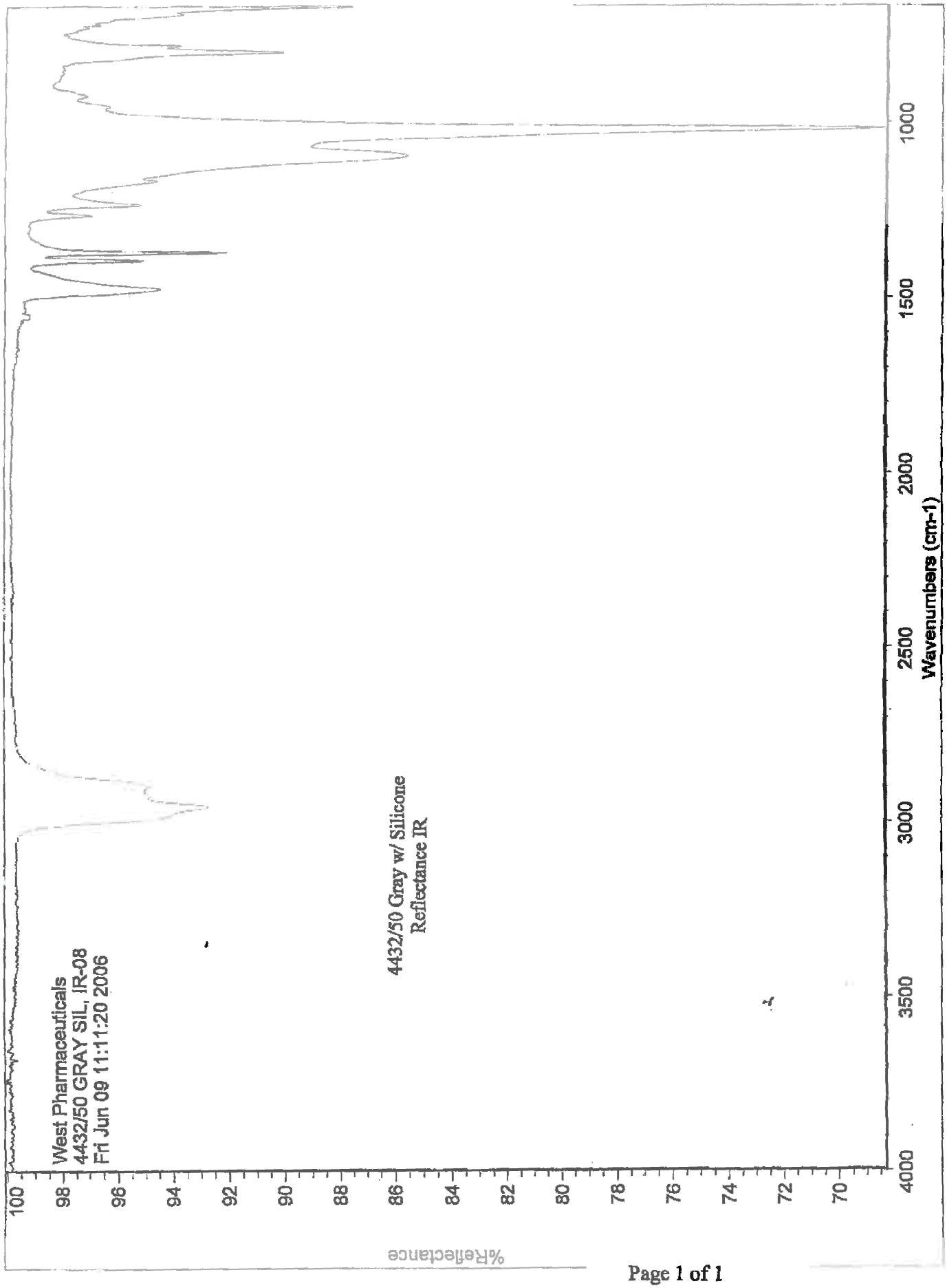


4432/50 Gray FC
Attachment 6

12 Feb 2013



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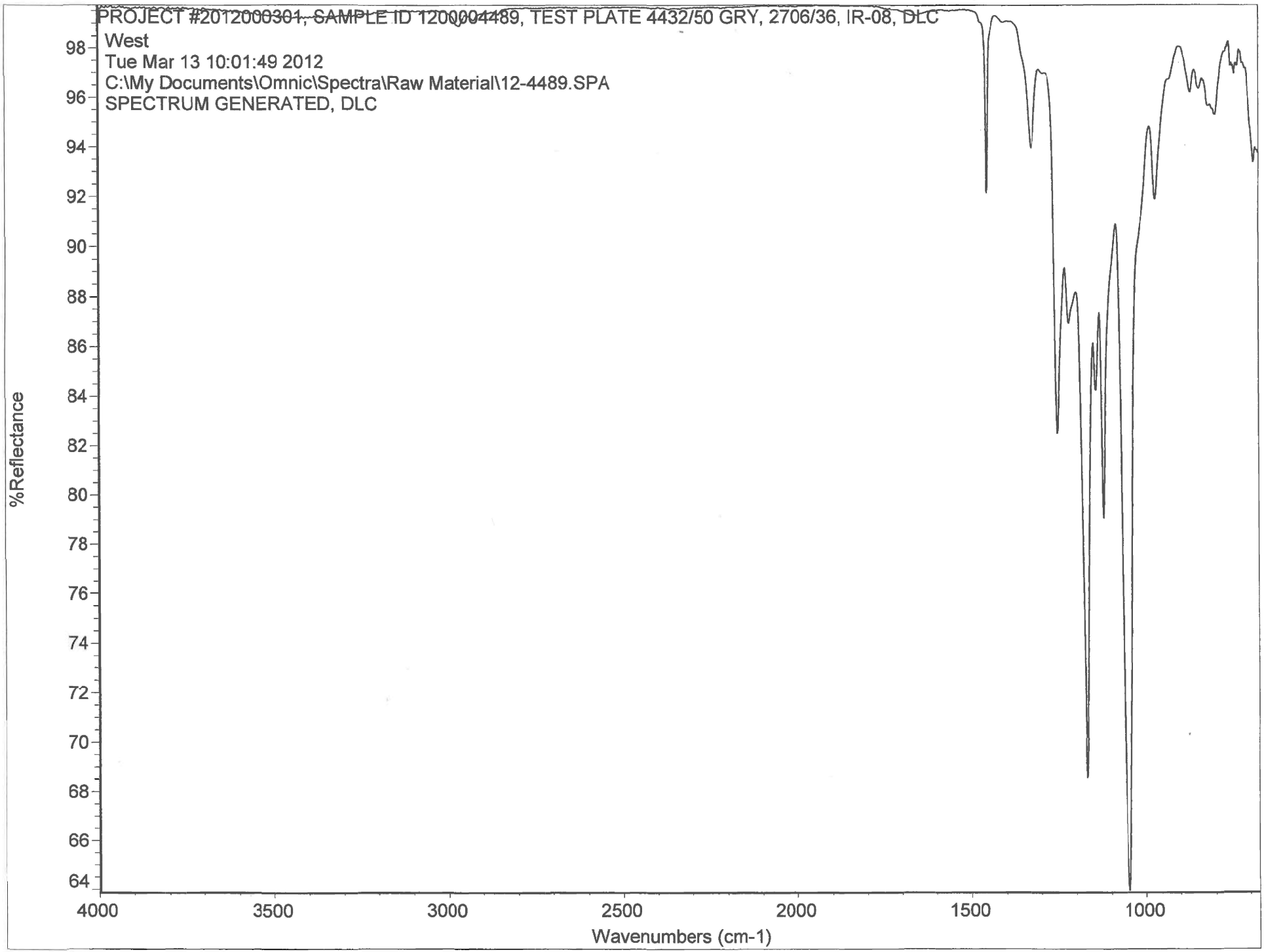
DLC 13 Mar 2012



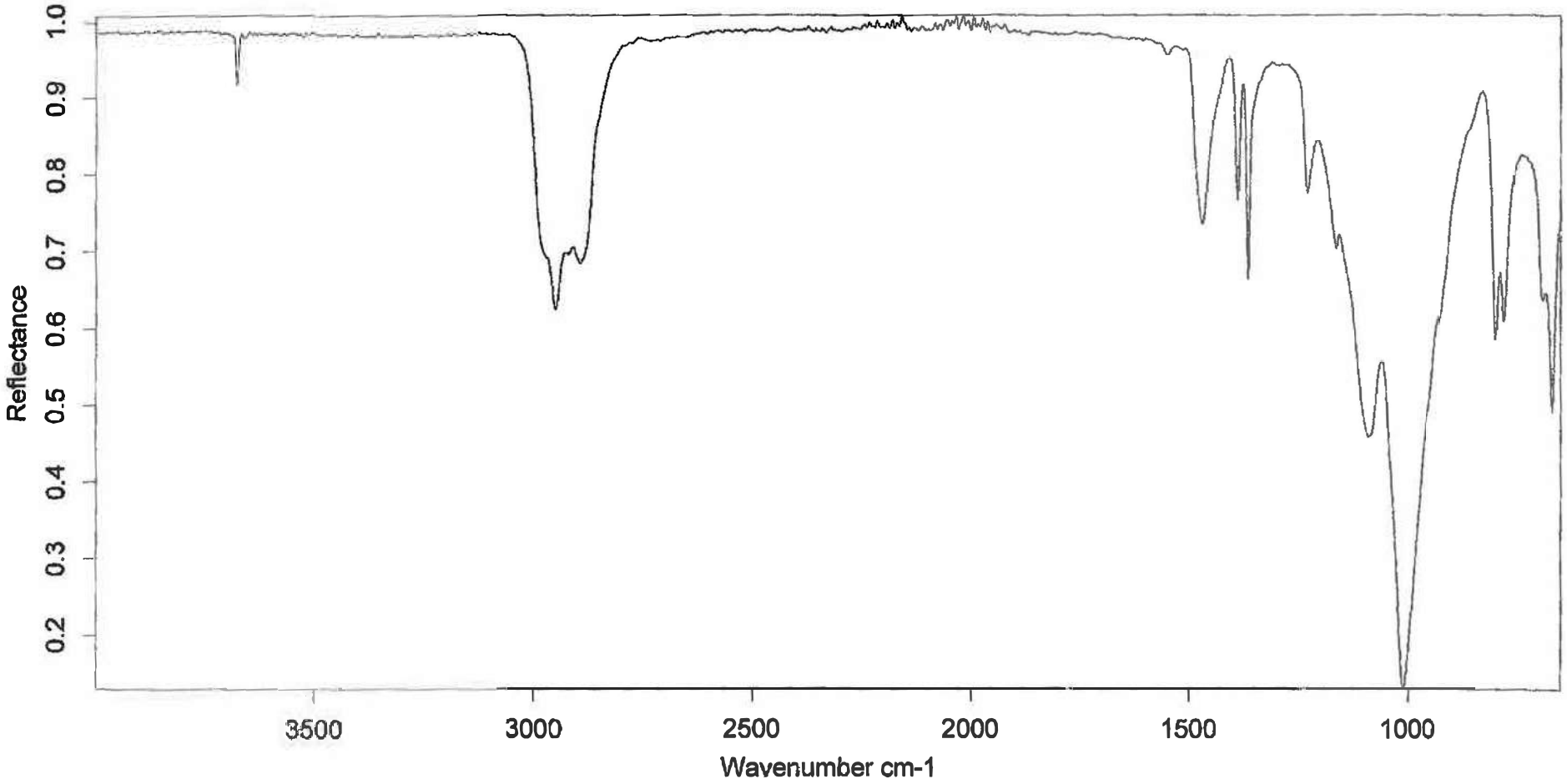
✓ Vhal 13 Mar 2012

4432/50 Gray FC
Attachment 8

West®
By your side
for a healthier world™

West Pharmaceutical Services	IR-26 WPS-US cut-surface
Deutschland GmbH&Co.KG	Operator: T. Beulen 08. OKT. 2014
EQS-AL; Analytical Laboratory	Collection time: 27/03/2014 15:34:45
	Scans: 12 | Gerät: 254 & 279

Typical IR-26 spectrum of formulation 4432-50-grey



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Attachment 9

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West

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for a healthier world™

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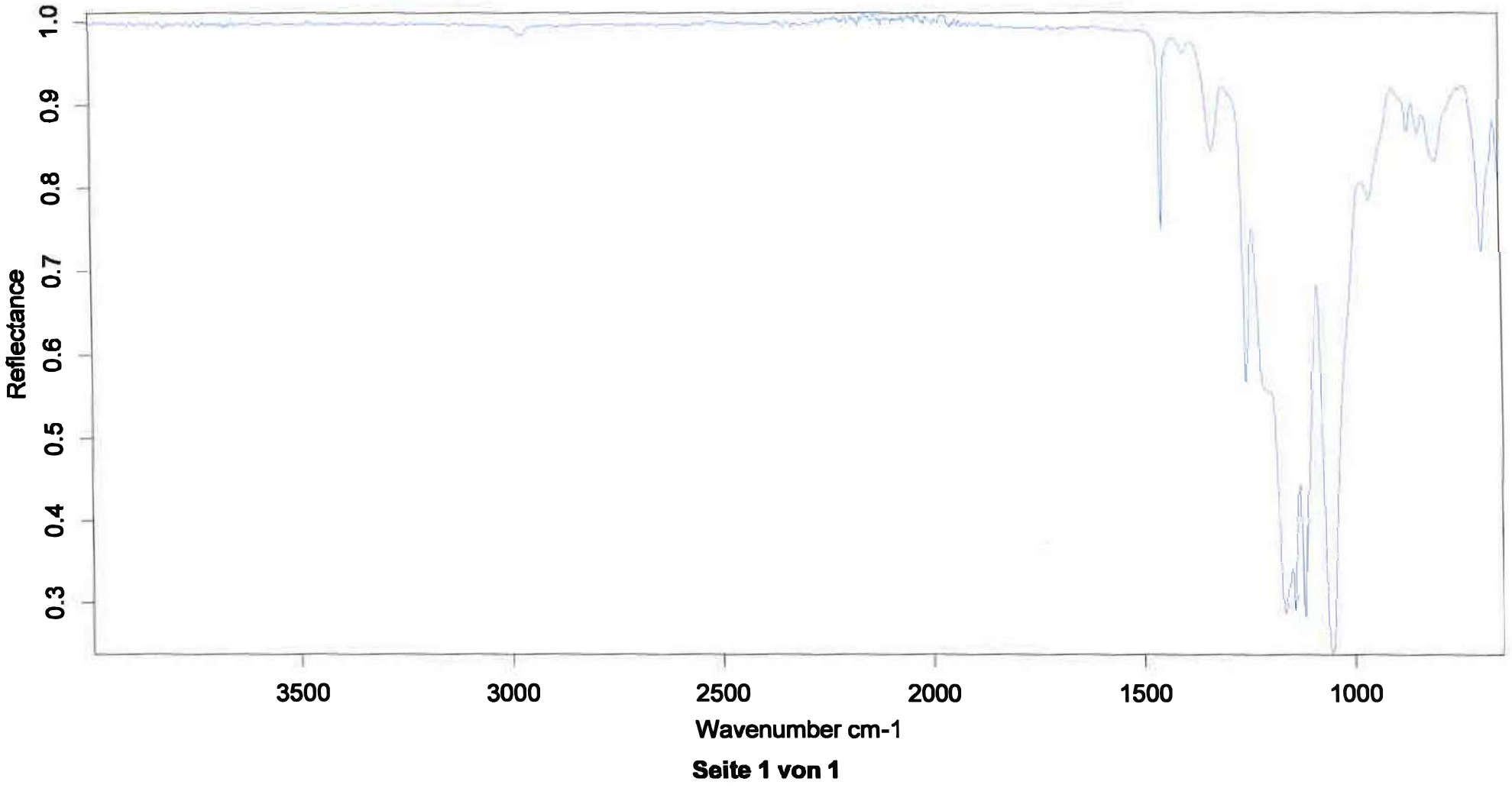
West Pharmaceutical Services	Operator: SI printed 30. APR. 2019
PPS Europe, Chemical Analytical Services	Collection Time: 15.06.2018 12:11:13
Alpha ATR platinum Diamond 1 Refl #9AE5522D	Instrument: 291 S/N: 1 02004
Number of Scans: 12	Resolution: 4

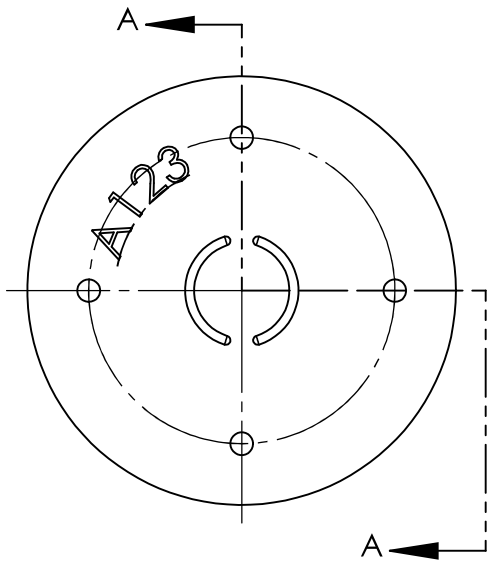
Typical ATR-spectrum of FluroTec

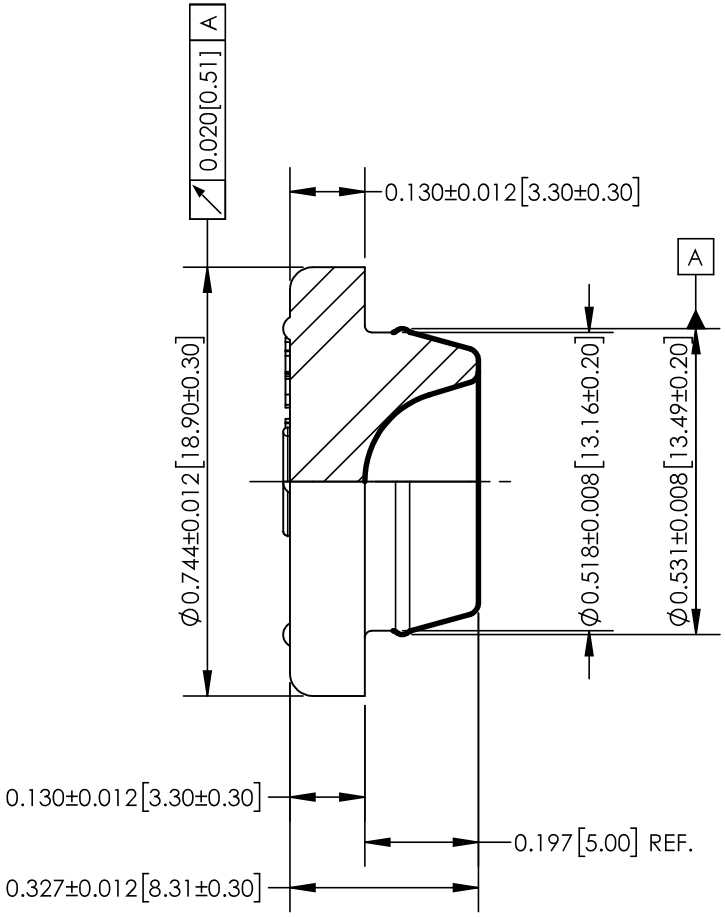


4432/50 Gray FC
Attachment 10

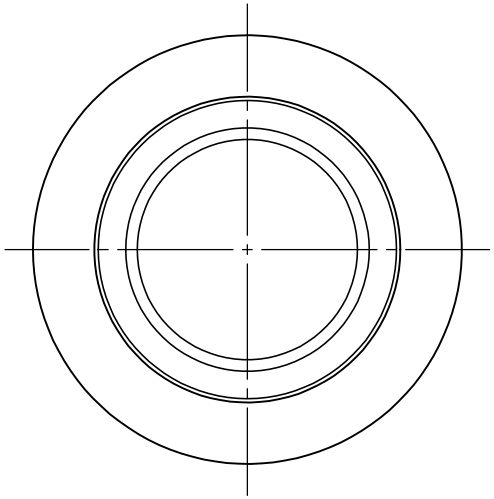
NOTES:
1. A MAX. OF 0.005" [0.13] RADIUS ALLOWABLE ON ALL SHARP CORNERS.
2. VALUES DENOTED BY [X.XX] ARE IN MILLIMETERS.

UNCONTROLLED COPY
Notification of changes will not be provided unless requested





SECTION A-A



ACTUAL SIZE

THICK LINE INDICATES LAMINATED FILM

REV.	DATE	DESCRIPTION	DR.	APP.
6	25 JUN 2019	PR# 60417	HG	DJS
5	20 JUL 2018	PR# 56848	ARD	RB
4	17 JUL 2018	PR# 56766	ARD	HG
3	10 JUL 2018	PR# 56661	ARD	HG
2	MAY05	ECN# 10690 (UPDATE TOLERANCES & REF. DIMENSION)	KN	MN
1	OCT03	ECN# 10553 (REVISION 1 IS FOR OFFICIAL RELEASE, ETC.)	CPC	JAR

REVISIONS

Reference Drawings

Nom S.A. (1.501) In.^2(9.68) Cm.^2
Nom Vol. (0.083) In.^3

TOLERANCES UNLESS NOTED

DECIMAL	METRIC
.XX ± .01	.X ± .3
.XXX ± .005	.XX ± .13
.XXXX ±	.XXX ±

FRACTIONAL: INCH: ± 1/64
ANGULAR: ± 1°
SURFACE FINISH: √

THE INFORMATION ON THIS DRAWING IS PROPRIETARY. THIS INFORMATION SHALL NOT BE DISCLOSED, REPRODUCED IN ANY FORM OR USED FOR ANY PURPOSE, WITHOUT PRIOR WRITTEN CONSENT OF WEST PHARMACEUTICAL SERVICES.

DO NOT SCALE DRAWING

TOOL CODE:

APPROVALS	DATE
DWN. C.P. Ching	08 JUN 00
CHK'D. C.Whitson	27 OCT 03
DWG. APPVD. J.Rodgers	29 OCT 03
TOOL ENGR. D.Kochel	29 OCT 03
PROJ. ENGR. D.Schinski	27 OCT 03
APPRD. N/A	

SAP Number

SIZE A

West

TITLE: **WPS-S10-F451**

MAT'L. **4432/50**

DRAWING AND REV. NO: **WS-791 Rev 6**

SCALE:3:1 | WEIGHT: LB. | SHEET 1 OF 1

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